



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd Razia Alangir Banu
Student Name	G. Rishitha
Roll Number	2520030101
Branch	CSE
Course Outcome	CO5 — Evaluate and implement advanced sorting algorithms — Merge Sort, Quick Sort (with pivot strategies), Heap Sort, Counting Sort, and Radix Sort — by empirically benchmarking their performance on real datasets and arguing trade-offs among time complexity, space usage, cache efficiency, and stability for different application contexts.

Case Study Topic: SkillBridge – Student Performance Ranking and Course Popularity Analysis Using Advanced Sorting Algorithms

Work Done Summary:

In this case study, we implemented advanced sorting algorithms — Merge Sort, Quick Sort, Heap Sort, Counting Sort, and Radix Sort — for SkillBridge’s online skill development and certification platform. The platform manages large volumes of student records, course enrollments, certification IDs, and performance data, requiring efficient sorting mechanisms for ranking, analytics, and report generation.

The implementation focused on sorting student performance records based on scores, ranking courses according to popularity, and organizing certification records efficiently. Merge Sort was used for stable sorting of student performance datasets, Quick Sort was applied for fast course popularity ranking using pivot strategies, Heap Sort was utilized to identify top-performing students, Counting Sort was implemented for sorting bounded assessment scores, and Radix Sort was used for sorting large certification IDs.

The system benchmarked each sorting algorithm on different datasets to analyze execution time, memory usage, and stability. This improved data organization, faster report generation, and efficient ranking systems within the SkillBridge platform.

Implementation Details:

1. Student performance record sorting using Merge Sort.
2. Course popularity ranking using Quick Sort with pivot strategies.
3. Top-performing student identification using Heap Sort.
4. Assessment score sorting using Counting Sort.
5. Certification ID sorting using Radix Sort.
6. Benchmarking sorting performance on real student datasets.
7. Comparison of execution time, space usage, and stability of sorting algorithms.
8. Analysis of cache efficiency and trade-offs across sorting methods.
9. Ranking system optimization for student analytics and reporting.

Result: Screenshots / Output:

```
PS C:\Users\DELL> javac SkillBridgeSorting.java
>> java SkillBridgeSorting
Original Student Scores:
85 92 78 65 99 88 73

Merge Sort:
65 73 78 85 88 92 99

Quick Sort:
65 73 78 85 88 92 99

Heap Sort:
65 73 78 85 88 92 99

Counting Sort:
65 73 78 85 88 92 99

Radix Sort:
65 73 78 85 88 92 99
PS C:\Users\DELL>
```

Result and complexity:

1. Merge Sort efficiently sorted student records while maintaining stability.
2. Quick Sort ranked courses quickly with optimized pivot selection.
3. Heap Sort successfully identified top-performing students.
4. Counting Sort efficiently handled bounded score datasets.
5. Radix Sort sorted large certification IDs effectively.
6. Benchmarking showed performance trade-offs across different datasets.
7. The system improved ranking accuracy and report generation speed.
 - Merge Sort: $O(n \log n)$
 - Quick Sort (Average): $O(n \log n)$
 - Quick Sort (Worst): $O(n^2)$
 - Heap Sort: $O(n \log n)$
 - Counting Sort: $O(n + k)$
 - Radix Sort: $O(nk)$

GitHub Link:

<https://github.com/rishi07-lab/DSA-2-CASE-STUDIES>

Note: The code and github link and screenshots have been uploaded in the above link . Please check it out there.

Student Signature	Faculty Signature
<u>Rishitha</u>	<u></u>

